

① Symmetries & the Poincaré Group

We've started to set up our quantum mechanical description of how we'll understand particle physics — we'll look for a Hamiltonian, understand the Hilbert space (set of particle states), and then dig through the dynamics with a technique such as perturbation theory.

Soon, we'll take this perturbative approach seriously and consider the Hilbert space of a free quantum field — this will give us the H_0 we can perturb around.

First, though, we'll pause to discuss the concept of symmetries. Symmetries are of foundational importance in particle physics, and will make our immediate goals easier by helping to organize our Hilbert space and the dynamics of the Hamiltonian.

A symmetry transformation is a change in the description of the system that does not alter the laws of physics, and thus, the results of experiments.

In quantum mechanics, the state of the system is described by a ray $\mathbb{R}$ in Hilbert space, which we take to be a collection of normalized vectors (satisfying $\langle \mathbb{R} | \mathbb{R} \rangle = 1$) w/ any two vectors

is the ray identical, up to a phase (i.e., $\mathcal{E}' = c\mathcal{E}$ for $|c|^2 = 1$ for $\mathcal{E}, \mathcal{E}' \in \mathbb{R}$.)

An observer $\mathcal{O}$ performs an experiment and finds the state represented by $\mathcal{R}$ changes to $\mathcal{R}_1$, or $\mathcal{R}_2$, or... w/ probability $P(\mathcal{R} \rightarrow \mathcal{R}_n)$.

Another observer $\mathcal{O}'$ who is related to the first observer by a symmetry transformation, will observe the system in a different state, $\mathcal{R}'$, with the result of the experiment being $\mathcal{R}'_1, \mathcal{R}'_2, \dots$ w/ probabilities $P(\mathcal{R}' \rightarrow \mathcal{R}'_n)$.

Since the two observers are related by a symmetry transformation, they must observe the same outcomes, meaning:

$$P(\mathcal{R} \rightarrow \mathcal{R}_n) = P(\mathcal{R}' \rightarrow \mathcal{R}'_n)$$

It was shown by E. Wigner that for any symmetry transformation T that takes rays $\mathcal{R} \rightarrow \mathcal{R}'$, one can define an operator U on Hilbert space that maps $\mathcal{E} \in \mathcal{R}$ to $\mathcal{E}' \equiv U\mathcal{E} \in \mathcal{R}'$,

with U being either unitary and linear or anti-unitary and antilinear.

More concretely, these possibilities are:

$$\langle U\Phi | U\Psi \rangle = \langle \Phi | \Psi \rangle \quad (\text{unitary})$$

$$+ U[\zeta|\Phi\rangle + \eta|\Psi\rangle] = \zeta U|\Phi\rangle + \eta U|\Psi\rangle \quad (\text{linear})$$

or

$$\langle U\Phi | U\Psi \rangle = \langle \Phi | \Psi \rangle^* \quad (\text{anti-unitary})$$

$$+ U[\zeta|\Phi\rangle + \eta|\Psi\rangle] = \zeta^* U|\Phi\rangle + \eta^* U|\Psi\rangle \quad (\text{anti-linear})$$

where ζ, η are complex numbers.

The adjoint of a linear operator L is defined as

$$\langle \Phi | L^\dagger \Psi \rangle = \langle L\Phi | \Psi \rangle$$

To show this is consistent with the linearity property, take the inner product of $L^\dagger(\zeta|\Phi_1\rangle + \eta|\Phi_2\rangle)$ w/ $|\Phi\rangle$:

$$\begin{aligned} \langle \Phi | L^\dagger(\zeta|\Phi_1\rangle + \eta|\Phi_2\rangle) &= \langle L\Phi | (\zeta|\Phi_1\rangle + \eta|\Phi_2\rangle) \\ &= \zeta \langle L\Phi | \Phi_1 \rangle + \eta \langle L\Phi | \Phi_2 \rangle \\ &= \langle \Phi | (\zeta L^\dagger|\Phi_1\rangle + \eta L^\dagger|\Phi_2\rangle) \end{aligned}$$

The unitarity ~~condition~~ condition & adjoint definition imply that $U^\dagger = U^{-1}$.

$$\begin{aligned} \langle U\Phi | U\Psi \rangle &= \langle U^\dagger U\Phi | \Psi \rangle = \langle \Phi | U^\dagger U\Psi \rangle \\ &= \langle \Phi | \Psi \rangle \end{aligned}$$

If we instead consider an anti-linear operator A , it's straightforward to see that the definition of the adjoint above, $\langle \Phi | A^\dagger \Psi \rangle = \langle A\Phi | \Psi \rangle$ is inconsistent w/ the anti-linear condition. Instead, the adjoint of A is defined as:

$$\langle \Phi | A^\dagger \Psi \rangle = \langle A\Phi | \Psi \rangle^* = \langle \Psi | A\Phi \rangle$$

(The last step is just a property of inner products)

To see this is consistent, again consider an inner product:

$$\begin{aligned} & \langle \Phi | A^\dagger (\xi |\Phi_1\rangle + \eta |\Phi_2\rangle) \rangle \\ &= [\langle A\Phi | (\xi |\Phi_1\rangle + \eta |\Phi_2\rangle)]^* \\ &= \xi^* \langle A\Phi | \Phi_1 \rangle^* + \eta^* \langle A\Phi | \Phi_2 \rangle^* \\ &= \xi^* \langle \Phi | A^\dagger \Phi_1 \rangle + \eta^* \langle \Phi | A^\dagger \Phi_2 \rangle \\ &= \langle \Phi | (\xi^* A^\dagger |\Phi_1\rangle + \eta^* A^\dagger |\Phi_2\rangle) \rangle \end{aligned}$$

The anti-unitary condition & adjoint defⁿ for A imply:

~~$$\langle A\Phi | A\Psi \rangle = \langle A^\dagger A\Phi | \Psi \rangle$$~~

$$\begin{aligned} \langle A\Phi | A\Psi \rangle &= \langle A^\dagger A\Phi | \Psi \rangle^* = \langle \Phi | \underbrace{A^\dagger A}_{\mathbb{1}} \Psi \rangle^* \\ &= \langle \Phi | \Psi \rangle^* \quad \implies \quad A^\dagger A = \mathbb{1} \end{aligned}$$

So for both unitary + linear or anti-unitary + anti-linear operators, we find

$$U^\dagger U = \mathbb{I} \text{ or } U^\dagger = U^{-1}$$

The identity transformation maps $\mathbb{R} \rightarrow \mathbb{R}$ and is thus a trivial symmetry transformation. It is represented on Hilbert space by the operator $U = \mathbb{I}$, which is clearly unitary + linear.

Any symmetry transformation that can be continuously deformed to the trivial one by a continuous change in parameters (e.g., a rotation or boost) must therefore be represented by a unitary, linear operator.

As a result, most symmetry transformations of interest to us in particle physics are represented by unitary, linear operators.

The one exception is time-reversal transformations, which are represented by anti-unitary, anti-linear operators.

Infinitesimal Symmetry Transformations

Let's now consider continuous transformations that can be deformed to the identity by a change of parameters*. In particular, consider a transformation that is infinitesimally close to the identity, represented by the unitary, linear operator:

$$U = 1 + i\epsilon t$$

* infinitesimal, real param.

The unitarity property requires:

$$(1 + i\epsilon t)^{\dagger} = (1 + i\epsilon t)^{-1}$$

$$\rightarrow 1 - i\epsilon t^{\dagger} \simeq 1 - i\epsilon t + \mathcal{O}(\epsilon^2)$$

$$\Rightarrow t^{\dagger} = t$$

That is, t must be Hermitian.

The linearity of U requires:

$$(1 + i\epsilon t)(\zeta|\Phi\rangle + \eta|\Psi\rangle) = \zeta(1 + i\epsilon t)|\Phi\rangle + \eta(1 + i\epsilon t)|\Psi\rangle$$

$$\rightarrow t(\zeta|\Phi\rangle + \eta|\Psi\rangle) = \zeta t|\Phi\rangle + \eta t|\Psi\rangle$$

i.e., t must also be linear.

Note that since $\hat{t}$ is a Hermitian operator, it has the potential to be an observable.

We'll see many examples where this is the case (e.g., energy, momentum, angular momentum, electric charge...)

A Hermitian operator can also be diagonalized on the Hilbert space: i.e., we can label our basis of states by their eigenvalue under $\hat{t}$.

This will be a very powerful tool for understanding particle states!

Group Theory

The set of symmetry transformations all have all the required properties to form a mathematical structure called a group.

We'll introduce some basic notions of group theory into our discussion as needed.

First, the definition: A group G is a set of elements $g \in G$ with a rule for multiplication s.t. for any ordered pair $(g, g') \in G$, the product $g \cdot g' \in G$ (closure.)

The group must satisfy three properties:

1) associativity: for every $g, g', g'' \in G$ the relation $(g \cdot g') \cdot g'' = g \cdot (g' \cdot g'')$ holds

2) identity: there exists an element e (or 1) such that $e \cdot g = g \cdot e = g$ for all $g \in G$.

3) inverse: for every element $g \in G$, there exists ~~an inverse~~ a unique element $g^{-1} \in G$ s.t.
 $g^{-1} \cdot g = g \cdot g^{-1} = e$.

One can easily reason out that the set of all symmetry transformations $T: \mathbb{R} \rightarrow \mathbb{R}$ form a group.

For any transformation T , we can define an operator $U(T)$ that acts on vectors $|\mathbb{R}\rangle$ in rays $\mathbb{R}$.

The operators $U(T)$ have properties that parallel the group structure, with one subtlety because $U(T)$ act on vectors in the Hilbert space.

In particular, if we carry out transformations in succession, T_1 & T_2 this maps a vector $|\mathbb{R}\rangle$ to $U(T_1)U(T_2)|\mathbb{R}\rangle$.

In contrast, we could carry out the transformation $T_2 T_1$ directly on $|\mathcal{E}\rangle$ to get the state $U[T_2 T_1]|\mathcal{E}\rangle$.

The two transformed vectors can differ by at most a phase:

$$U[T_2] U[T_1] = \exp(i\phi(T_2, T_1)) U[T_2 T_1]$$

If the phase $\phi = 0$, we say the operators form a representation of the group.

If ϕ is non-vanishing, the operators $U(T)$ provide a so-called projective representation.

Projective representations are very important in quantum mechanics, and appear in connection w/ both the rotation group and the larger Lorentz group.

Fortunately, a symmetry group w/ projective representations can always be enlarged, without changing the physical predictions, in such a way that all of its representations can be defined as non-projective, with $\phi = 0$.

For now, we will assume this has been done, and take the composition rule to be

$$U[T_2] U[T_1] = U[T_2 T_1]$$

Lie Groups

We now consider in more detail a very important class of symmetry transformations that form a special kind of group, known as a Lie Group.

Lie groups describe transformations $T(\theta)$ that are specified by a finite number of real, continuous parameters, θ^a , w/ each group element connected to the identity by a path within the group.

For two successive transformations, $T(\theta) \neq T(\bar{\theta})$, the group multiplication rule is given by

$$T(\bar{\theta}) T(\theta) = T(f(\theta, \bar{\theta}))$$

with ~~for~~ some function $f(\theta, \bar{\theta})$. Here, θ collectively denotes all parameters θ^a . It is possible to define $T(\theta)$ in such a way that $\theta = 0$ corresponds to the identity, i.e., ~~$T(0) = \mathbb{I}$~~ $T(0) = \mathbb{I}$. Plugging this into the group multiplication rule, we have:

$$T(\bar{\theta}) \mathbb{I} = T(\bar{\theta}) = T(f(\bar{\theta}, 0))$$

$$\mathbb{I} T(\theta) = T(\theta) = T(f(0, \theta))$$

Thus we require for each θ^a :

$$f^a(\theta, 0) = f^a(0, \theta) = \theta^a.$$

For small values of the parameters $\theta, \bar{\theta}$ we can expand $f^a(\theta, \bar{\theta})$ near the identity to second order:

Consistency w/ the condition $f^a(\theta, 0) = f^a(0, \bar{\theta}) = \theta^a$ gives

$$f^a(\theta, \bar{\theta}) = \theta^a + \bar{\theta}^a + \theta^b \bar{\theta}^c f^a_{bc} + \dots$$

Let's now consider the unitary operators on Hilbert space corresponding to these symmetry transformations.

We can represent these operators near the identity as

$$U(T(\theta)) = \mathbb{I} + i\theta^a t_a + \frac{1}{2}\theta^a \theta^b t_{bc} + \dots$$

where t_a, t_{bc} are Hermitian operators, indep. of the θ^a parameters. Note that $t_{bc} = t_{cb}$.

The operator composition rule for ordinary (non-projective) representations is

$$U(T(\bar{\theta})) U(T(\theta)) = U(T(f(\bar{\theta}, \theta)))$$

We now expand this in powers of $\theta^a, \bar{\theta}^a$, using our above expansion for f :

$$\begin{aligned} & \left[\mathbb{I} + i\bar{\theta}^a t_a + \frac{1}{2}\bar{\theta}^b \bar{\theta}^c t_{bc} + \dots \right] \left[\mathbb{I} + i\theta^d t_d + \frac{1}{2}\theta^e \theta^f t_{ef} + \dots \right] \\ &= \mathbb{I} + i(\theta^a + \bar{\theta}^a + f^a_{bc} \bar{\theta}^b \theta^c + \dots) t_a \\ & \quad + \frac{1}{2}(\theta^b + \bar{\theta}^b + \dots)(\theta^c + \bar{\theta}^c + \dots) t_{bc} + \dots \end{aligned}$$

The terms of order $\mathbb{1}$, A , $\bar{\theta}$, θ^2 , $\bar{\theta}^2$ on each side cancel off.

However, the $\bar{\theta}\theta$ terms give a non-trivial condition:

$$(i\bar{\theta}^b t_b)(i\theta^c t_c) = if^a{}_{bc} \bar{\theta}^b \theta^c t_a + \frac{1}{2}(\bar{\theta}^b \bar{\theta}^c + \theta^c \bar{\theta}^b) t_{bc}$$

$$\hookrightarrow -t_b t_c = if^a{}_{bc} t_a + t_{bc}$$

or,

$$t_{bc} = -t_b t_c - if^a{}_{bc} t_a$$

In the steps above, we've freely relabelled dummy indices and used $t_{bc} = t_{cb}$. The equation above shows that if we know the operator appearing at first order in A in the expansion of $U(T(\theta))$

and also the quadratic coefficient $f^a{}_{bc}$ that appears in $f(\bar{\theta}, \theta)$, which follows from the structure of the gauge, then we can calculate the second order terms in $U(T(\theta))$.

Because t_{bc} is symmetric, ($t_{bc} = t_{cb}$) we have:

$$-t_b t_c - i f^a_{bc} t^a = -t_c t_b - i f^a_{cb} t_a$$

$$\Rightarrow t_b t_c - t_c t_b = i(-f^a_{bc} + f^a_{cb}) t_a$$

or,

~~t_b, t_c~~

$$\boxed{[t_b, t_c] = i C^a_{bc} t_a}$$

where we've defined $C^a_{bc} = -f^a_{bc} + f^a_{cb}$. The Hermitian operators t^a are known as generators of the group, while the set of constants C^a_{bc} are called the structure constants.

The boxed equation describes a set of commutation relations respected by the generators of the group, known as the Lie algebra.

Given the generators & structure constants, one can continue the process above to determine the complete power series for $U(T(A))$, at least for the group elements connected to the identity,

A generic group element of this type may be represented ~~by~~ in terms of the matrix exponential:

$$U[T(\theta)] = \exp(i\theta^a t_a)$$

Abelian groups

If the structure constants C^a_{bc} vanish, then all the generators of the group commute.

$$[t_a, t_b] = 0$$

This is the case of an Abelian group.

Since the generators all commute, the order in which the transformation is performed is not important, and the unitary operators $U[T(\theta)]$ all commute as well.

The Lorentz & Poincaré Groups

We have already met the Lorentz group in a fashion, now we'll return to it w/ more formalism & consider the full Poincaré Group.

The basic principles of relativity can be formulated starting from a coordinate transformation.

If $x^\mu = (t, \vec{x})$ are the coords. in one inertial frame & we consider a (symmetry!) transf. to another frame w/ coords. x'^μ , the basic fact of SR is that the line elements must be equivalent:

$$ds^2 = g_{\mu\nu} dx'^\mu dx'^\nu = g_{\mu\nu} dx^\mu dx^\nu$$

w/ $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. Equivalently, we want

$$g_{\mu\nu} \frac{\partial x'^\mu}{\partial x^\rho} \frac{\partial x'^\nu}{\partial x^\sigma} = g_{\rho\sigma}$$

The most general transf. that satisfies this is an inhomogeneous Lorentz transf. or a Poincaré transformation!

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu$$

Here we have parameters:

a^μ : space + time translations

Λ^μ_ν : Lorentz transf. (rotations, boosts, reflections)

We already know Λ must satisfy

$$g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = g_{\rho\sigma} \quad (*)$$

The Poincaré transformations form a group.

If we consider two successive transf.:

$x^\mu \rightarrow x'^\mu$ defined by (Λ, a) then $x'^\mu \rightarrow x''^\mu$ defined by $(\bar{\Lambda}, \bar{a})$, we have:

$$\begin{aligned} x''^\mu &= \bar{\Lambda}^\mu_\rho x'^\rho + \bar{a}^\mu \\ &= \bar{\Lambda}^\mu_\rho (\Lambda^\rho_\nu x^\nu + a^\rho) + \bar{a}^\mu \\ &= (\bar{\Lambda}^\mu_\rho \Lambda^\rho_\nu) x^\nu + (\bar{\Lambda}^\mu_\rho a^\rho + \bar{a}^\mu) \end{aligned}$$

$\rightarrow$ transform. defined by $(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a})$.

Trivial to check ~~$\bar{\Lambda}\Lambda$~~ $\bar{\Lambda}\Lambda$ satisfies $(*)$, and $\bar{\Lambda}a + \bar{a}$ is still a constant 4-vector

$\rightarrow$ satisfy the closure requirement of multiplication rule.

$$T(\bar{\Lambda}, \bar{a}) T(\Lambda, a) = T(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a})$$

We already know each Λ^μ_ν has an inverse $(\Lambda^{-1})^\rho_\nu$

$$(\Lambda^{-1})^\rho_\nu \equiv \Lambda_\nu^\rho = g_{\nu\mu} g^{\rho\sigma} \Lambda^\mu_\sigma$$

→ Inverse of Poincaré transf. $T(\Lambda, a)$ is $T(\Lambda^{-1}, -\Lambda^{-1}a)$.

$$\begin{aligned} T(\Lambda^{-1}, -\Lambda^{-1}a) T(\Lambda, a) &= T(\Lambda^{-1}\Lambda, \Lambda^{-1}a + (-\Lambda^{-1}a)) \\ &= T(1, 0) \quad \text{the identity.} \end{aligned}$$

So the transformations $T(\Lambda, a)$ form a group.
(the Poincaré Group)

It has a number of important subgroups
(subset of elements of G that themselves satisfy the group axioms)

- the homogeneous Lorentz group (set $a=0$)

$$T(\bar{\Lambda}, 0) T(\Lambda, 0) = T(\bar{\Lambda}\Lambda, 0)$$

→ the group $SO(1, 3)$

(some properties worked out in Hw)

- The group of pure translations: $\mathbb{R}_4$

$$T(1, a) T(1, \bar{a}) = T(1, a + \bar{a})$$

→ this is an Abelian subgroup.

- The subgroup of proper, orthochronous Lorentz transformations: $(SO(1, 3))^+$

$$T(\Lambda, 0) \quad \text{w/} \quad \det \Lambda = +1, \quad \Lambda^0_0 \geq +1$$

→ set of transf. continuously connected to the identity, preserved under multiplication.

Any element of the homogeneous Lorentz group is obtained by an element of $SO(1, 3)^+$ composed w/ P, T or PT .

Now we wish to understand how the Poincaré group structure is reflected in operators on Hilbert space.

$$|E\rangle \rightarrow U(1, a)|E\rangle.$$

$\hat{U}$ unitary operators on Hilbert space.

The Poincaré Group has both ordinary + projective representations.

We'll discuss ~~how~~ later how to enlarge the Poincaré group so that all reps. are non-projective.

Assuming this has been done, the $U(1, a)$ satisfy:

$$U(\bar{\lambda}, \bar{a}) U(\lambda, a) = U(\lambda \bar{\lambda}, \lambda a + \bar{a})$$

as operators.

The enlargement of the Poincaré group later will lead us to spinors (2 fermions).

The Poincaré Algebra

We'll learn a lot about the structure of the Poincaré group by looking at its Lie Algebra.

To start, consider the ~~the~~ set of infinitesimal transf.:
The identity is given by

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \text{~~0~~} \quad (a^\mu = 0)$$

So ~~a transf.~~ an infinitesimal transformation can be written as

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \underset{\substack{\uparrow \\ \text{infinitesimal parameters}}}{\omega^\mu_\nu}, \quad a^\mu = \underset{\substack{\uparrow \\ \text{infinitesimal parameters}}}{\epsilon^\mu}$$

The Lorentz transf. condition on Λ : $\Lambda^T g \Lambda = g$
~~tells~~ tells us at leading order:

$$\begin{aligned} g_{\mu\nu} (\delta^\mu_\rho + \omega^\mu_\rho) (\delta^\nu_\sigma + \omega^\nu_\sigma) &= g_{\rho\sigma} \\ &= \cancel{g_{\rho\sigma}} + \omega_{\rho\sigma} + \omega_{\sigma\rho} = \cancel{g_{\rho\sigma}} + \mathcal{O}(\omega^2) \end{aligned}$$

$$\rightarrow \omega_{\sigma\rho} + \omega_{\rho\sigma} = 0 \quad : \quad \omega_{\mu\nu} \text{ is } \underline{\text{antisymmetric}}$$

(note this is a condition on ω w/ lowered indices!)

So an infinitesimal Poincaré transf. is given by specifying $\omega^\mu{}_\nu$ & a^μ w/ $\omega_{\mu\nu} = -\omega_{\nu\mu}$

An infinitesimal rank four tensor has 6 indep. components (3 rotations, 3 boosts) + 4 translations
 $\rightarrow$ total of 10 parameters.

Now we turn to the Unitary operator $U(1+\omega, e)$, which can be written:

$$U(1+\omega, e) = 1 - \frac{1}{2}i\omega_{\rho\sigma} J^{\rho\sigma} - i\varepsilon_\rho P^\rho + \dots$$

(We've added some $1/2$, -1 's for later convenience — this is just a definition!)

Since $\omega_{\rho\sigma}$ is anti-symmetric, we take $J^{\rho\sigma}$ to be antisymmetric as well (its symmetric component would vanish in the product anyhow!) : $J^{\rho\sigma} = -J^{\sigma\rho}$

$\Rightarrow$ 10 Hermitian operators $P^\mu, J^{\mu\nu}$

P^0 — energy operator (the Hamiltonian!)

P^i — spatial ~~translation~~ momenta

J^{ij} — ~~rotation~~ angular momenta

How do these operators transform under a Lorentz transformation?

Under $|\Phi\rangle \rightarrow U|\Phi\rangle$ for a unitary transf.,
a state w/ an operator t acting on it transf. as:

$$\begin{aligned} t|\Phi\rangle &\rightarrow U(t|\Phi\rangle) \\ &= U t U^{-1} U|\Phi\rangle \end{aligned}$$

In other words, we have ~~$t \rightarrow U t U^{-1}$~~

$$|\Phi\rangle \rightarrow U|\Phi\rangle \text{ along with } t \rightarrow U t U^{-1}$$

for all operators t acting on Hilbert space.

So, let's consider this product; ~~first for $2p^{\mu}$~~

$$U(\Lambda, a) \cdot U(1+\omega, \epsilon) U^{-1}(\Lambda, a)$$

Just as for the transformation itself, we have for the unitary operators:

$$U^{-1}(\Lambda, a) = U(\Lambda^{-1}, -\Lambda^{-1}a),$$

so we can apply the general rule for composition w/
 $\bar{\Lambda} = \Lambda^{-1}$, $\bar{a} = -\Lambda^{-1}a$.

We find:

$$\begin{aligned} U(\Lambda, a) &\star U(1+\omega, \epsilon) \star U^{-1}(\Lambda, a) \\ &= U(\Lambda(1+\omega)\Lambda^{-1}, \Lambda\epsilon - \Lambda\omega\Lambda^{-1}a). \end{aligned}$$

Since $\Lambda \mathbb{1} \Lambda^{-1} = \mathbb{1}$,

$$= U(1 + \Lambda\omega\Lambda^{-1}, \Lambda\epsilon - \Lambda\omega\Lambda^{-1}a)$$

Now we can expand this to first order in ϵ and ω , yielding:

$$\begin{aligned} U(\Lambda, a) &\left[\mathbb{1} - \frac{1}{2}i\omega_{\rho\sigma} J^{\rho\sigma} - i\epsilon_{\rho} P^{\rho} \right] U^{-1}(\Lambda, a) \\ &= \mathbb{1} - \frac{1}{2}i \left[\Lambda\omega\Lambda^{-1} \right]_{\mu\nu} J^{\mu\nu} - i \left[\Lambda\epsilon - \Lambda\epsilon\Lambda^{-1}a \right]_{\mu} P^{\mu} \end{aligned}$$

Since $U \mathbb{1} U^{-1} = \mathbb{1}$, the identity cancels on each side.

Next looking at the Lorentz transformation $(\Lambda\omega\Lambda^{-1})_{\mu\nu}$, let's first ~~take~~ consider $(\Lambda\omega\Lambda^{-1})^{\mu}_{\nu}$, since this was the structure we originally gave ω^{μ}_{ν} .

$$(\Lambda\omega\Lambda^{-1})^{\mu}_{\nu} = \Lambda^{\mu}_{\rho} \omega^{\rho}_{\sigma} (\Lambda^{-1})^{\sigma}_{\nu} = \Lambda^{\mu}_{\rho} \Lambda_{\nu}^{\sigma} \omega^{\rho}_{\sigma}$$

then lowering the index,

$$\begin{aligned} (\Lambda\omega\Lambda^{-1})_{\mu\nu} &= g_{\mu\tau} \Lambda^{\tau}_{\rho} \Lambda_{\nu}^{\sigma} g^{\rho\tau} \omega_{\tau\sigma} \\ &= (g_{\mu\tau} g^{\tau\rho} \Lambda^{\tau}_{\rho}) \Lambda_{\nu}^{\sigma} \omega_{\tau\sigma} = \Lambda_{\mu}^{\tau} \Lambda_{\nu}^{\sigma} \omega_{\tau\sigma} \end{aligned}$$

Note that this is exactly what we could have inferred from the position of the indices — both are down, so ~~are~~ both Λ matrices appear as the inverse.

So, cancelling some factors of i , ± 1 , we have

$$\begin{aligned} U(\Lambda, a) \left[-\frac{1}{2} \omega_{\rho\sigma} J^{\rho\sigma} - \epsilon_\rho P^\rho \right] U^{-1}(\Lambda, a) \\ = -\frac{1}{2} (\Lambda \omega \Lambda^{-1})_{\mu\nu} J^{\mu\nu} - (\Lambda \epsilon - \Lambda \omega \Lambda^{-1})_\mu P^\mu \end{aligned}$$

First, consider this equation w/ $\omega_{\rho\sigma} = 0$. Then,

$$\begin{aligned} U(\Lambda, a) [-\epsilon_\rho P^\rho] U^{-1}(\Lambda, a) &= -(\Lambda \epsilon)_\mu P^\mu \\ \Rightarrow -\epsilon_\rho U(\Lambda, a) P^\rho U^{-1}(\Lambda, a) &= -\epsilon_\rho (\Lambda^{-1})^\rho{}_\mu P^\mu \\ &= -\Lambda_\mu{}^\rho \epsilon_\rho P^\mu \end{aligned}$$

So we learn that the operator P^ρ transforms as

$$U(\Lambda, a) P^\rho U^{-1}(\Lambda, a) = \Lambda_\mu{}^\rho P^\mu \quad (*)$$

→ This is the transformation we'd expect for a 4-vector, except that it has Λ^{-1} , rather than Λ .

Nevertheless, this is a promising result for a Hermitian operator that corresponds to an observable related to momentum.

Next we take $\epsilon_p = 0$, $\omega_{\rho\sigma} \neq 0$. With a little algebra, following similar steps, we find:

$$U(\Lambda, a) J^{\rho\sigma} U^{-1}(\Lambda, a) = \Lambda_{\mu}^{\rho} \Lambda_{\nu}^{\sigma} \left[\overset{\uparrow}{J^{\mu\nu}} - a^{\nu} P^{\mu} + a^{\mu} P^{\nu} \right] \quad (**)$$

a transformed angular momentum
 the new component of $\vec{J}$
 from changing the axis of
 rotation

→ again, this looks promising for our interpretation of $J^{\mu\nu}$ as a "4-angular momentum" tensor.

Now we can consider how P^{μ} , $J^{\mu\nu}$ transform under an infinitesimal Poincare transformation.

We take

$$\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu} + \omega^{\mu}_{\nu} ; \quad a^{\mu} = \epsilon^{\mu}$$

w/ ω & ϵ infinitesimal params.

(Note these are different than the dummy params we used to derive $U P U^{-1} \approx U J U^{-1}$).

To first order in ω , the inverse Lorentz transformation is:

$$(\Lambda^{-1})^\rho_\mu = \Lambda^\rho_\mu = \delta^\rho_\mu + \omega^\rho_\mu$$

as one can check:

$$\begin{aligned} (\Lambda^{-1})^\rho_\mu \Lambda^\mu_\nu &= (\delta^\rho_\mu + \omega^\rho_\mu)(\delta^\mu_\nu + \omega^\mu_\nu) \\ &= \delta^\rho_\nu + \omega_\nu^\rho + \omega^\rho_\nu + O(\omega^2) \\ &= \delta^\rho_\nu + g^{\rho\sigma} \underbrace{(\omega_{\nu\sigma} + \omega_{\sigma\nu})}_{=0 \text{ — antisymmetry}} + O(\omega^2) \\ &= \delta^\rho_\nu. \end{aligned}$$

Now applying the results of (*) & (**), we find first for from the transf. of P^μ :

$$U(\Lambda, a) P^\rho U^{-1}(\Lambda, a) = \Lambda^\rho_\mu P^\mu$$

$$\begin{aligned} \Rightarrow \left(1 - \frac{1}{2}i\omega_{\mu\nu} J^{\mu\nu} - i\epsilon_\mu P^\mu\right) P^\rho \left(1 + \frac{1}{2}i\omega_{\rho\sigma} J^{\sigma\rho} + i\epsilon_\rho P^\rho\right) \\ = (\delta^\rho_\mu + \omega^\rho_\mu) P^\mu \end{aligned}$$

$$\Rightarrow P^\rho - i\left[\frac{1}{2}\omega_{\mu\nu} J^{\mu\nu} + \epsilon_\mu P^\mu, P^\rho\right] + \dots = P^\rho + \omega^\rho_\mu P^\mu$$

where we've rewritten the ~~first~~ products as a commutator.
difference in

~~$$\Rightarrow 1 - i\left[\frac{1}{2}\omega_{\mu\nu} J^{\mu\nu} + \epsilon_\mu P^\mu, P^\rho\right] + \dots$$~~

Cancelling terms, we find

$$-i \left[\frac{1}{2} \omega_{\mu\nu} J^{\mu\nu} + \epsilon_\mu P^\mu, P^\rho \right] = \omega_\mu{}^\rho P^\mu \quad (1)$$

Next, the transformation of $J^{\mu\nu}$ yields' (**):

$$\begin{aligned} -i \left[\frac{1}{2} \omega_{\mu\nu} J^{\mu\nu} + \epsilon_\mu P^\mu, J^{\rho\sigma} \right] \\ = \omega_\mu{}^\rho J^{\mu\sigma} + \omega_\nu{}^\sigma J^{\rho\nu} - \epsilon^\sigma P^\rho + \epsilon^\rho P^\sigma \end{aligned} \quad (2)$$

(I'm leaving out ~ a page of simple algebra, similar in spirit to the above)

Now we equate the coefficients of $\omega_{\mu\nu}$ & ϵ_μ on each side of (1) & (2):

(since $\omega_{\mu\nu}$, ϵ_μ were arbitrary infinitesimals, their coefficients must individually match)

From (2):

$$-i \frac{1}{2} \omega_{\mu\nu} [J^{\mu\nu}, J^{\rho\sigma}] = \omega_\mu{}^\rho J^{\mu\sigma} + \omega_\nu{}^\sigma J^{\rho\nu}$$

$$\begin{aligned} \rightarrow \omega_{\mu\nu} [J^{\mu\nu}, J^{\rho\sigma}] &= i(2g^{\rho\nu} \omega_{\mu\nu} J^{\mu\sigma} + 2g^{\sigma\mu} \omega_{\mu\nu} J^{\rho\nu}) \\ &= -\omega_{\mu\nu} \end{aligned}$$

Splitting terms & relabelling indices, we get:

$$\omega_{\mu\nu} [J^{\mu\nu}, J^{\rho\sigma}] = i \left[g^{\rho\nu} \omega_{\mu\nu} J^{\mu\sigma} + g^{\rho\mu} \omega_{\mu\nu} J^{\nu\sigma} \right. \\ \left. - g^{\sigma\mu} \omega_{\mu\nu} J^{\rho\nu} - g^{\sigma\nu} \omega_{\mu\nu} J^{\rho\mu} \right]$$

$\Rightarrow$

$$[J^{\mu\nu}, J^{\rho\sigma}] = i (g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\sigma\mu} J^{\rho\nu} + g^{\sigma\nu} J^{\rho\mu})$$

Similarly, the coefficient of $\epsilon_{\mu\nu}$ in (2) gives:

$$-i\epsilon_{\mu\nu} [P^\mu, J^{\rho\sigma}] = -\epsilon^\sigma P^\rho + \epsilon^\rho P^\sigma$$

$$\Rightarrow \epsilon_{\mu\nu} [P^\mu, J^{\rho\sigma}] = i (g^{\rho\mu} \epsilon_{\mu\nu} P^\sigma - g^{\sigma\mu} \epsilon_{\mu\nu} P^\rho)$$

$\Rightarrow$

$$[P^\mu, J^{\rho\sigma}] = i (g^{\rho\mu} P^\sigma - g^{\sigma\mu} P^\rho)$$

The coefficient of $\epsilon_{\mu\nu}$ in (1) tells us:

$$[P^\mu, P^\nu] = 0$$

$\rightarrow$ These expressions together are the Lie Algebra of the Poincaré group, aka. the Poincaré Algebra.

Summarizing: the general set of transformations consistent w/ preserving the line element are generated by the Hermitian operators P^μ , $J^{\mu\nu}$ satisfying the Lie Algebra commutation relations:

$$[P^\mu, P^\nu] = 0, \quad [P^\mu, J^{\rho\sigma}] = i(g^{\mu\rho}P^\sigma - g^{\mu\sigma}P^\rho)$$

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\sigma\mu}J^{\rho\nu} + g^{\sigma\nu}J^{\rho\mu})$$

P^μ is the momentum generator, w/ P^0 = energy or the Hamiltonian, P_i the spatial momentum generator.

$\vec{J} = (J^{23}, J^{31}, J^{12})$ is the angular momentum generator

$\vec{K} = (K^1, K^2, K^3) = (J^{10}, J^{20}, J^{30})$ are the generators of boosts.

Note that $\vec{P}$ & $\vec{J}$ commute w/ P^0 , the Hamiltonian

→ momentum & angular momentum are conserved!
(the eigenvalues are good labels for states)

Boosts don't commute, & are not conserved.

We can rewrite the commutation relations in 3-dimensional notation:

$$\underbrace{[J^i, J^j] = i\epsilon^{ijk} J^k}_{\text{su}(2)} \quad [J^i, K^j] = i\epsilon^{ijk} K^k$$

$$[K^i, K^j] = -i\epsilon^{ijk} J^k \quad [J^i, P^j] = i\epsilon^{ijk} P^k$$

$$[K^i, P^j] = i\delta^{ij} H \quad [K^i, H] = iP^i$$

$$[J^i, H] = [P^i, H] = [H, H] = [P^i, P^j] = 0$$

(ϵ^{ijk} is totally antisymmetric, $\epsilon^{123} = +1$)

As a final note, just as the $\text{su}(2)$ algebra lets us build ~~up the Hilbert space~~ construct the representations w/ different spins to build a (finite-dimensional) Hilbert space of a qubit, we can use the Poincaré algebra

to build up representations of the Poincaré group
 $\rightarrow$ these are the particles that build the Hilbert space for particle physics!

I'll include separate notes on this derivation, but we'll instead get our particle states from QFT.